Block Chain

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A Blockchain-Based Reward Mechanism for Mobile Crowdsensing

I. Abstract

Mobile crowd Sensing (MCS) is a new sensor scenario for cyber physical social systems. MCS is widely used in smart cities, personal healthcare and environmental monitoring. The MCS application retrieves sensory data from the target area by recruiting participants and assigning rewards. Reward mechanisms are important for encouraging participants to participate and providing sensory data. However, while MCS applications perform reward mechanisms, sensory data and personal information can be at great risk to malicious task initiators / participants and hackers. This article proposes a new blockchain-based MCS framework that protects both the capture process and the incentive mechanism by protecting privacy and leveraging new blockchain technology. To provide a fair incentive mechanism, this article considered the MCS scenario as a sensory data market where the market classifies participants into two categories: monthly and immediate payment participants. This article proposes an incentive mechanism supported by a 3-level Stackelberg game by analyzing two different types of participants and task initiators. Through theoretical analysis and simulation, evaluation addresses two aspects. Reward mechanism and blockchain-based MCS performance. The proposed reward mechanism improves the usefulness of task starters by up to 10% compared to traditional Stackelberg games. You can also maintain the market share required for monthly fee participants while achieving sustainable sensory data delivery. An assessment of the blockchain-based MCS has shown that as the number of participants increases, the delay increases tolerably. Finally, this article discusses the future challenges of blockchain-based MCS.

Introduction

The development of network technology, sensor devices, and social networks has facilitated the rollout of mobile crowd sensing (MCS), the next-generation Internet of Things (IoT). MCS is a new sensor framework equipped with a smartphone sensor and incorporating human intelligence into a loop. MCS is a typical application of Cyber-Physical and Social Systems (CPSS) with an interdisciplinary approach that brings together knowledge from communications, computer science, computer networks, economics, psychology, and social research to provide solutions for sensor tasks. became. .. Compared to traditional IoT frameworks, MCS has advantages such as wide detection range, spatiotemporal sensor data, deployment feasibility, and flexibility. Challenges studied at MCS include incentive mechanisms, sensory data transmission, etc

II. Literature Survey

Junqin Huang et.al [1] "mobile crowd sensing Based on Blockchain used in industrial systems" Smart factories are a key element in transforming traditional computerized industries into data-driven smart industries, but the cost-effectiveness and reliability of intelligent industrial systems. Gender, mobility, and scalability cannot be achieved. Data-driven industrial systems rely primarily on sensor data collected from statically deployed sensors. However, the spatial coverage of industrial sensor networks is limited due to high deployment and maintenance costs. Recently, Mobile crowd Sensing (MCS) has emerged as a new sensor paradigm with benefits such as cost efficiency, mobility, and scalability. However, due to its centralized architecture, traditional MCS systems are vulnerable to malicious attacks and single points of failure. To this end, MCS will be integrated into the industrial system without the need for additional dedicated devices. To overcome the shortcomings of conventional MCS systems, we propose a blockchain-based MCS system (BMCS). Specifically, we use miners to review sensory data and design incentive mechanisms for dynamic reward ranking (DRR) to reduce imbalances in multiple sensory tasks. Along with that, we are also developing a sensory data quality detection scheme that will identify and mitigate data anomalies. Here we Implement a BMCS prototype on Ethereum after that we perform extensive experiments in a realistic factory workspace. Both experimental results and security analysis show that BMCS can protect industrial systems and improve system reliability.

Tassos Dimitriou [2] "Fair and Private Bitcoin Rewards: Incentives to Participate in Cloud Sensing Applications" This task develops a rewarding framework that can be used as a component of cloud sensing applications. Our protocol allows users to send data and receive Bitcoin payments in a way that protects their privacy, preventing vendors who like it from tying data and payments to users. At the same time, it prevents malicious user behavior, such as duplicate redemption attempts by users to earn rewards by sending the same data multiple times. More importantly, ensuring the fairness of the exchange.

Peter Danielis, et.al [3"Urban Count: Mobile Crowd Counting in Urban Environments" This task introduces Urban Count, a fully distributed crowd counting protocol for high-density cities. UrbanCount relies on mobile device-to-device communication to perform bulk quotes. Each node collects mass size estimates from other participants in the system and immediately integrates these estimates into local estimates whenever they are within range. The goal of UrbanCount is to create an accurate mapping of local estimates and expected global results while preserving node privacy. Evaluate the proposed protocol through extensive track-driven simulations of synthetic and realistic mobility models. In addition, we investigated the accuracy and density dependencies and showed that the difference in local estimates in a dense environment was less than 2% in synthetic scenarios and less than 7% in realistic scenarios.

Phuong Nguyen et.al [4] "Context-aware crowd sensing in opportunistic mobile social networks". In this paper, we investigate physical cloud sensing problems and connect them to vertex cover problems in graph theory. Finding the optimal solution for the vertex cover problem is NP-complete, and known approximation algorithms do not work well in cloud sensing scenarios, so we suggest the terms node observability and coverage utility score find vertex coverage. Design a new context-aware approximation algorithm Customized for cloud sensing tasks. In addition, we design a human-centric bootstrap strategy for the initial mapping of sensory devices within the physical set based on social information about the user (interests, friendships, etc.). Experiments with real-world data traces show that the proposed approach far exceeds the baseline approximation algorithm in terms of acquisition range.

Taewan You et.al [5] "Mobile crowd sensing based on CICN". In This article, we propose a mobile crowd sensing application that is based on Community Information-Centric Networking to collect opportunistic sensor data in a limited area where the radius of the Bluetooth Low Energy beacon is limited. This application has more valuable features compared to regular mobile crowd-sensing applications. This application will also support data integrity and use a simple communication model for IPless communication via common CICN features. Then collect the sensor data using a name that includes the BLE beacon identifier. This application concentrates more on user privacy and data integrity.

Leye Wang et.al [6"A new mobile crowd sensing framework for energy-efficient and cost-effective data uploads". This article tells about effSense. It is a power-efficient and low-cost data upload framework that uses adaptive upload schemes within a fixed data upload cycle. At each cycle, effSense provides participants with a decentralized decision-making scheme to choose the right time and network to upload data. effSense reduces data costs for NDP users by maximally offloading data to Bluetooth / WiFi gateways or DP users encountered. It will reduce the power consumption by using piggybacking data on the call By leveraging call predictability and user mobility, effSense chooses the right upload strategy for both user types. According to MIT's reality mining and Nodobo dataset analysis, effSense reduces DP user power consumption by 55% to 65% and NDP user data cost by 48% to 52% compared to traditional upload schemes. Reduce.
Haiming Jin, et.al [7"Quality Awareness Incentive Mechanisms for Mobile crowd Sensing" This article explains about some key issues with MCS systems like incentives for worker participation etc. Unlike existing work, we propose an incentive framework for the MCS system called Thanos, which contains an important indicator called Worker Information Quality (QoI). Due to various factors (sensor quality, ambient noise, etc.), the quality of sensory data provided by individual workers varies significantly. It is not a big effort for the MCS platform to obtain high-quality data. Technically, Thanos' design is based on a reverse combinatorial auction. Consider a combination auction model for both single-minded and multi-minded. For the former, we design a true, individually rational, and computationally efficient mechanism that guarantees near-optimal social welfare. after that, we design a repetitive descent mechanism that meets individual rationality and computational efficiency and guarantees close ratios.

Dimitris Chatzopoulos et.al [8] "Privacy protection and cost-optimal mobile cloud sensing using smart contracts on the blockchain" In this work, mobile cloud sensing uses blockchain technology and smart contracts for the spatial cloud sensing required by mobile users. The best way we suggest coordinating is the interaction between the provider and the mobile user at a specific location to perform the task. Smart contracts, which act as a process running on the blockchain, are used to protect user privacy and make payments. In addition design a true, cost-effective auction to assign cloud sensing tasks to mobile users. This minimizes payments from crowdsensing providers to mobile users. Extensive experimental results show that the proposed privacy auction outperforms the current proposal in terms of the cost of large numbers of mobile users and tasks.

Objectives

This article is aimed at a blockchain-based MCS that enables anonymization of participant IDs, decentralized reward allocation, and transparent transactions without the usual trusted third parties.

Existing System

Crowdsourcing systems that use human intelligence to solve complex tasks have received considerable interest and acceptance in recent years. However, most existing crowdsourcing systems rely on centralized servers that are affected by the weaknesses of traditional trust-based models, such as single points of failure. It is also vulnerable to distributed denial of service (DDoS) and Sybil attacks due to the involvement of malicious users. In addition, the high service fees of crowdsourcing platforms can hinder the development of crowdsourcing. How to deal with these potential problems is scientific and of considerable value.

Problem Definition

(1) Unreliable participants: Participants who may falsify their identities or reputations due to enemy attacks. Malicious participants can steal sensory data, resulting in privacy breaches. (2) Unreliable Task Initiator: The task initiator may publish the recognition task without guarantee of reward and attempt to steal personal information from the participant when communication is made between the participants.(3) High system operating costs: The MCS must have the authority to handle all communications between the initiator and participant of the collection task. Given that the MCS architecture is usually centralized, MCS can be added with a single point of failure.

Proposed System:

This article proposes a blockchain-based MCS that enables the anonymization of participant IDs, decentralized reward allocation, and transparent transactions without the usual trusted third parties. This article consists of three main contributions. (1) Proposal of a blockchain-based MCS framework that provides participant privacy protection, secure detection process, and reward allocation mechanism by utilizing new blockchain technology. (2) Design a blockchain-based MCS workflow and a set of smart contracts that support the automation of MCS discovery task execution. Once all the participants' identities have been validated on the blockchain, the steps to perform the recognition task are triggered. Once the task initiator collects sensory data, the reward allocation procedure begins to run. Smart contracts can guarantee the automation and security of the MCS framework. (3) Study the sensory data market and the characteristics of participants. This article divides participants into three parts that are monthly paying participants, immediate paying participants, and task initiators. It provides an economic approach for analyzing incentive mechanisms. By leveraging the three-tier Stackelberg game reward mechanism, MCS can enable a fair and efficient market for sensory data. Benefits: (1) As a result, monthly paying participants are assured that sensory data will be provided in the long term. (2) Sustainable perception It is also possible to maintain the market share required for monthly payment participants while realizing data distribution.

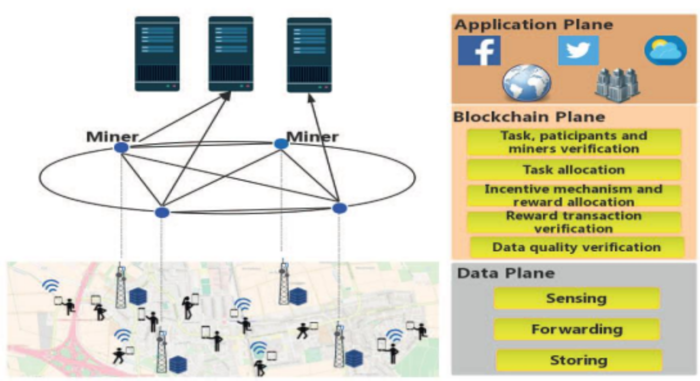
Modules

(1) Task Initiator: An initiator that publishes recognition tasks and assigns rewards to participants with monthly and immediate payments via the blockchain. (2) Participants: Participants are divided into two different roles, which are participants who are paid immediately after completing the sensor task and participants who are paid monthly. Instant pay participants' will get rewards based on the quality and reputation of sensory data, and Monthpay participants' rewards are salaries based on the number of completed tasks and reputation. (3) Miners: Adding authorized miners is intended to confirm the identities of all participants and transactions between participants. In this scenario, the hybrid base station acts as a certified miner on the blockchain-based MCS. Hybrid base stations can

not only carry out communication but also serve as storage and computing resources. By using the blockchain, task initiators, participants and miners work anonymously on the blockchain. In addition, authorized miners also verify the identity of the task's initiator and participant before continuing with the MCS collection task. Smart contracts are deployed to miners to perform discovery tasks. Since the miner stores all blocks in memory, it is responsible for verifying the quality control of task initiator and participant registrations, transactions, and sensory data. Figure 1 shows the architecture of the blockchain-based MCS. This article proposes a blockchain-based MCS three-tier architecture consisting of a data tier, a blockchain tier, and an application tier. Basic functions can be handled by smartphone users such as data collection, transfer, and storage at the data level. This architecture has an additional abstraction layer called the blockchain layer to help MCS applications identify participants, assign recognition tasks/rewards, execute transactions, and control the quality of sensory data. increase

The application layer can handle the demands of a particular organization and process sensory data to extract knowledge.

In this article, the hybrid base station is equipped with a server that can handle communication and computing tasks. The miner stores all blockchain information locally. Figure 2 shows the blockchain and block structure. The miner's chain begins with the Genesis block. Each block consists of the blockhead, the hash before the block, the timestamp, the state root, the root of the transaction, and so on. The state root and transaction root are the roots of the Merkel tree. The transaction root stores all hash values of transactions between participants. State routes include contract hashes, which are hashes for smart contracts, account balances, and storage routes.



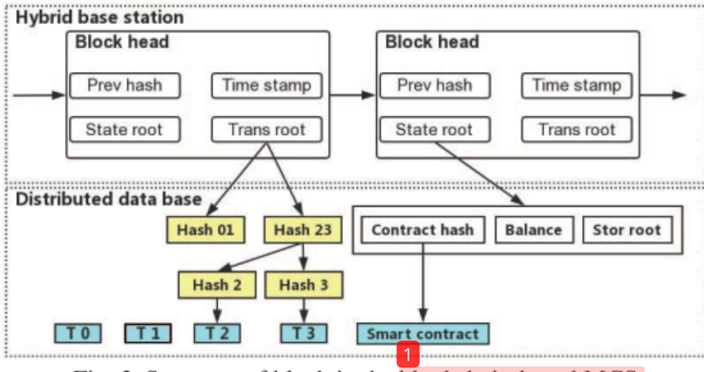


Fig. 2. Structure of block in the blockchain-based MCS.

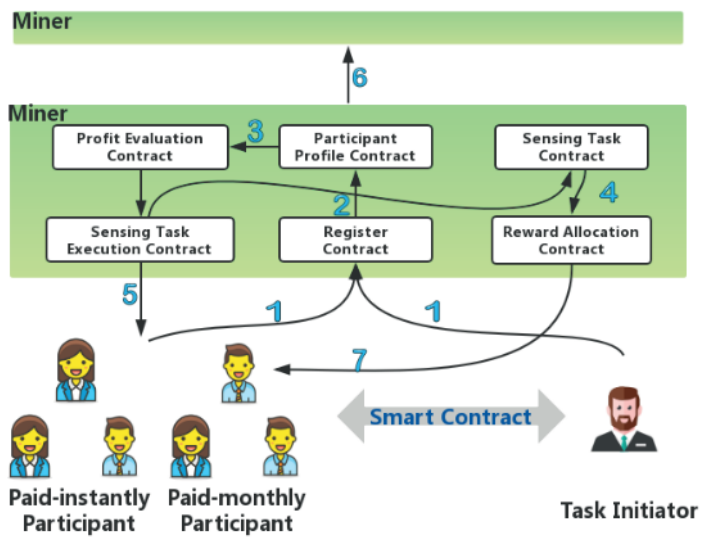


Fig. 3. Workflow of the blockchain-based MCS.

This section tells about the blockchain-based MCS workflow. Figure 3 shows a smart contract workflow between each entity in a blockchain-based MCS. The task initiator communicates with participants via a set of smart contracts deployed to miners. It is assumed that the participant (including the task initiator) has registered and registered with the Certificate Authority (CA) and after registering, they have received the required cryptographic material used for authentication at the start of the discovery task. (1) System initialization: MCS application task initiator and participant registration. Send your ID, public / private key, certificate, etc. to the nearest miner. The miner executes a "registration contract" and confirms the identities of participants and task initiators with other miners through a consensus mechanism. Then, if the participant's ID is registered, the miner sends a confirmation to the task initiator. after the completion of the registration process, the next process is triggered. The task initiator sends a description of the sensing task to the appropriate miner. The miner confirms the recognition task and then sends the recognition task to all registered participants. (2) Deployment of incentive mechanism (3-stage Stackelberg game): After system

initialization, the role of each ID becomes clear. because of that, the incentive mechanism is processed by the execution of the "Participant Profile Contract" and the "Capture Task Contract". This procedure is described in detail in the next section.
(3) Token allocation: after completion of the incentive mechanism all participants will receive a reward message and the size of the sensory data. You need to verify that the message is valid and then perform the discovery task according to the reward. Each participant is assigned a token that represents the perceptual task and reward. (4) Upload sensory data: Each participant will definitely get a reward after completing the sensor task, so upload the promised sensory data to the task initiator via the miner. (5) Generate a new block: The miner processes the proof of work (PoW) and builds a new block using all transactions of the capture task in the chain. The new block will be checked and eventually added to the blockchain. (6) Token Redemption and Task Completion: After adding all the blocks, the token is given to all the participants, and once the recognition task is complete, the tokens will be redeemed whenever and wherever they need them.
A smart contract means a contract that directs how each party acts when they trust each other. All smart contracts are assumed to be approved prior to deployment.

The proposed blockchain-based MCS framework is designed with a new set of smart contracts for manipulating transactions and validation. By Following the blockchain-based MCS workflow, we get some sets of smart contracts which are. (1) Registration Agreement: All participants, including the Task Initiator Registry, enter into a registration agreement as shown in Table I. All participants verify their identities by sending the miner their address and role in a secure communication channel. After the agreement between the miners is over, participants will be admitted without a legal signature. This article omits the technical details of the blockchain encryption algorithm. It employs an asymmetric encryption algorithm to provide a secure communication channel. (2) Participant Profile Contract: When the miner collects all information from a participant, a profile is created to support the participant selection process, detection task execution process, and reward awarding process. As shown in Table II, subscriber profiles include subscriber reputation, expected rewards for completing tasks, subscriber status, and more. (3) Sensing task contract: A sensing task is a contract that consists of some sensing tasks like sensing ID, execution status, deposit, and bounty plan. The execution status of the collection task is represented using a binary variable. A status of 0 means that the discovery task has not finished yet and vice versa. There are also parameters such as B. A reward plan that provides guidance on cognitive task deposits and reward allocations that can guarantee the promised rewards to participants.

(4) Profit Valuation Contract: When a participant registers with the chain, the miner executes the profit valuation contract to obtain all the information for valuing the reward allocation plans. I have an evaluation contract. The detection scheme in Table IV shows U () as a participant's profit function. (5) Contract to perform a perceptual task: When a participant calculates the maximum profit according to details such as perceptual task, reward, etc., it is as follows. A perceptual plan that includes the quality of the sensory data and the size of the sensory data. Participants perform perceptual tasks according to the perceptual task contracts defined in Table V. Therefore, the sensing task execution contract is triggered. (6) Reward Allocation Contracts: The main algorithms in this article are expanded to collection task contracts according to subscriber profile contracts. The capture task's reward plan is returned by the algorithm. Meanwhile, during this process, rewards are

assigned to participants as tokens and are activated by the reward allocation contract in Table VI.

TABLE I

REGISTRATION CONTRACT

ID	Address	Type
P_1	Addr$\{P_1\}$	Initiator
P_2	Addr$\{P_2\}$	Instant-pay participant
P_3	Addr$\{P_3\}$	Monthly-pay participant
P_4	Addr$\{P_4\}$	Miner

TABLE II

PARTICIPANT PROFILE CONTRACT

Address	Profile	Sensing Task ID	Reward of Task
Addr$\{P_1\}$	Profile$\{P_1\}$	T_1	R_1
Addr$\{P_2\}$	Profile$\{P_2\}$	t_1	r_1
Addr$\{P_3\}$	Profile$\{P_3\}$	t_2	r_2
Addr$\{P_4\}$	Profile$\{P_4\}$	t_2	0

TABLE III

SENSING TASK CONTRACT

Sensing Task ID	Status	Deposit	Reward Plan
T_1	0	D_1	R_1
T_2	1	D_2	R_2

TABLE IV

PROFIT EVALUATION CONTRACT

ID	Expecting reward	Sensing task ID	Device ability	Profit $U(\cdot)$
P_1	r_1	T_1	$D(P_1)$	$U_1(\cdot)$
P_2	r_2	T_2	$D(P_2)$	$U_2(\cdot)$

TABLE V

SENSING TASK EXECUTION CONTRACT

ID	Profit $U(\cdot)$	Sensing task ID	Status
P_1	$U_1(\cdot)$	T_1	0
P_2	$U_2(\cdot)$	T_2	1

TABLE VI

REWARD ALLOCATION CONTRACT

Sensing Task ID	Status	Participant's ID	Token
T_1	0	P_1	$Token_1$
T_2	1	P_2	$Token_2$

III. References

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